

DBMS Unit 4 Question Bank — Transactions & Concurrency Control

Database Management Systems (DBMS) · Unit 4 · Transactions & Concurrency Control · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Illustrate ACID Properties with a worked example.
2. What is meant by Serializability? Discuss its advantages and limitations.
3. Illustrate Two-Phase Locking with a worked example.
4. Discuss the role of Timestamp Ordering in Transactions & Concurrency Control.
5. What is meant by Deadlock Handling in Databases? Discuss its advantages and limitations.
6. Explain Isolation Levels with a neat diagram and a suitable example.

Part B — Long answer (10 marks)

1. Compare ACID Properties with Isolation Levels. When would you prefer one over the other?
2. What is meant by Serializability? Discuss its advantages and limitations.
3. What is meant by Two-Phase Locking? Discuss its advantages and limitations.
4. What is meant by Timestamp Ordering? Discuss its advantages and limitations.
5. Compare Deadlock Handling in Databases with ACID Properties. When would you prefer one over the other?
6. Compare Isolation Levels with Deadlock Handling in Databases. When would you prefer one over the other?
7. Illustrate ACID Properties with a worked example.
8. What is meant by Serializability? Discuss its advantages and limitations.
9. Compare Two-Phase Locking with ACID Properties. When would you prefer one over the other?
10. Compare Timestamp Ordering with Deadlock Handling in Databases. When would you prefer one over the other?
11. Illustrate Deadlock Handling in Databases with a worked example.
12. Explain Isolation Levels with a neat diagram and a suitable example.